

Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime behavior, and deployment.

1. System Requirements

System Name: ResuFit: Precision CV Matching System

Architecture Style: Microservices

Functional Requirements

- Submit CV
- View and Edit CV
- Answer Screening Questions
- Get Rating for CV
- Parse CV
- Compare CV with Job Requirements
- Validate Applicant Responses
- Generate Percentage-Based Result

Non-Functional Requirements

- Performance

Description: The system **MUST** respond to user requests within a reasonable time frame.

- Reliability

Description: The system **MUST** be able to recover from failures and maintain data integrity.

- Security

Description: The system **MUST** protect user data and prevent unauthorized access.

- Privacy

Description: The system **MUST** ensure that user data is handled in accordance with applicable laws and regulations.

- Scalability

Description: The system **MUST** be able to handle a large number of users and requests without compromising performance.

- Usability

Description: The system **MUST** provide an intuitive and user-friendly interface that is easy to navigate.

- Maintainability

Description: The system MUST be designed to be easily maintained and updated.

- Compatibility

Description: The system MUST be compatible with a variety of devices and browsers.

- Integration

Description: The system MUST be able to integrate with other systems and applications.

2. Formal Architecture Specification (ACME ADL)

The following section provides a formal architectural specification using ACME ADL, capturing components, components,

```
System ResuFit:PrecisionCVMatchingSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Microservices";
  Property source : String = "AI";
  Component CVParser = new Component {
    Port in;
    Port out;
    Property responsibility : String = "['Parse CV', 'Extract key information such as skills and years']";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component ScreeningQuestionManager = new Component {
    Port in;
    Port out;
    Property responsibility : String = "['Present screening questions to applicants', 'Validate applicant responses']";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component RatingGenerator = new Component {
    Port in;
    Port out;
    Property responsibility : String = "['Calculate CV ranking score', 'Calculate screening question scores']";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component CVComparator = new Component {
    Port in;
    Port out;
    Property responsibility : String = "['Compare CV with job requirements', 'Assess alignment between CV and job']";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component ChatGPTValidator = new Component {
    Port in;
    Port out;
    Property responsibility : String = "['Verify applicant responses', 'Grade applicant responses']";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
}
```

```

}
Component UserInterface = new Component {
    Port in;
    Port out;
    Property responsibility : String = "['Allow applicants to submit CVs', 'Allow applicants to view
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Attachment CVParser.out to CVComparator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment CVParser.out to RatingGenerator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment CVComparator.out to RatingGenerator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment ScreeningQuestionManager.out to RatingGenerator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment ChatGPTValidator.out to RatingGenerator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment UserInterface.out to CVParser.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment UserInterface.out to ScreeningQuestionManager.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment UserInterface.out to ChatGPTValidator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment UserInterface.out to RatingGenerator.in {
    Property delivery : String = "at-least-once";
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}
Attachment UserInterface.out to CVComparator.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
}

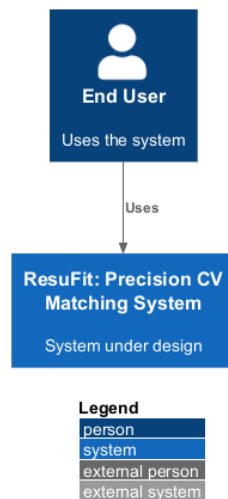
```

```

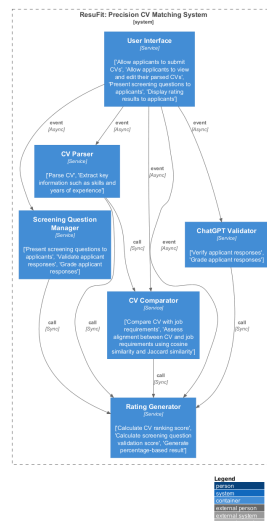
Property resilience_pattern : String = "Retry, CircuitBreaker";
Property delivery : String = "at-least-once";
Property ordering : Boolean = false;
Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
}

```

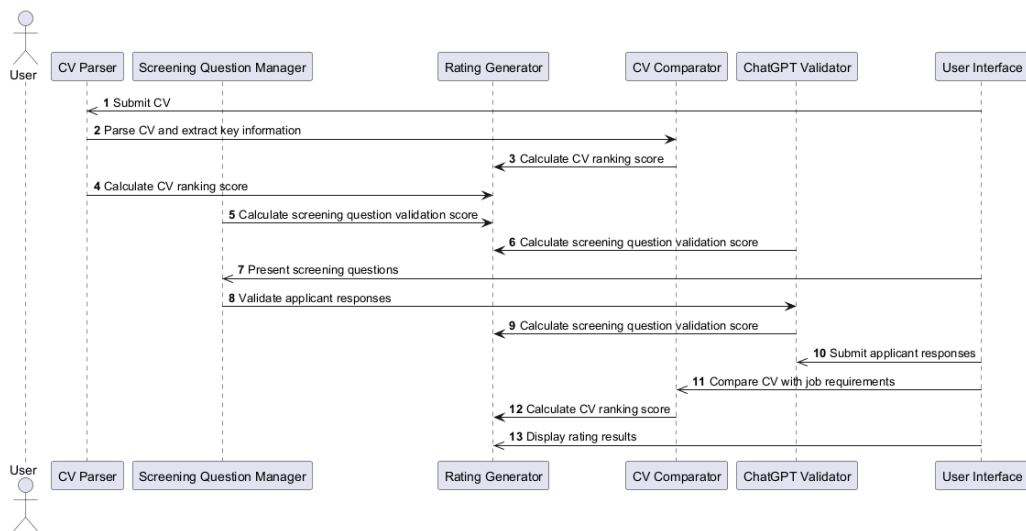
3. Context View



4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)

